

Tutorial 5: CS 215

October 22, 2025

1. Consider a class whose items are drawn from a multivariate Gaussian distribution with mean vector $\boldsymbol{\mu}_1 \in \mathbb{R}^d$ and covariance matrix $\mathbf{C}_1$ of size $d \times d$. Consider another class whose items are drawn from a multivariate Gaussian distribution with mean vector $\boldsymbol{\mu}_2$ and covariance matrix $\mathbf{C}_2$. Let the prior probability of the two classes be π_1 and π_2 respectively. How will you characterize the decision boundary between these two classes? That is for what values of $\mathbf{x} \in \mathbb{R}^d$, does it have equal probability of belonging to either of these classes? What happens further if both the covariance matrices are diagonal?

Solution:

Consider the two classes:

$$\begin{aligned}\text{Class 1: } \mathbf{X} &\sim \mathcal{N}(\boldsymbol{\mu}_1, \mathbf{C}_1), & P(\text{Class 1}) &= \pi_1, \\ \text{Class 2: } \mathbf{X} &\sim \mathcal{N}(\boldsymbol{\mu}_2, \mathbf{C}_2), & P(\text{Class 2}) &= \pi_2.\end{aligned}$$

The Bayes decision boundary is defined as the set of points $\mathbf{x}$ where the posterior probabilities are equal:

$$P(\text{Class 1} \mid \mathbf{X} = \mathbf{x}) = P(\text{Class 2} \mid \mathbf{X} = \mathbf{x}),$$

which is equivalent to

$$\pi_1 f_1(\mathbf{x}) = \pi_2 f_2(\mathbf{x}),$$

where $f_i(\mathbf{x})$ is the multivariate Gaussian PDF:

$$f_i(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\mathbf{C}_i|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^\top \mathbf{C}_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right], \quad i = 1, 2.$$

Taking logarithms, we obtain:

$$\ln \pi_1 - \frac{1}{2} \ln |\mathbf{C}_1| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_1)^\top \mathbf{C}_1^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) = \ln \pi_2 - \frac{1}{2} \ln |\mathbf{C}_2| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_2)^\top \mathbf{C}_2^{-1} (\mathbf{x} - \boldsymbol{\mu}_2).$$

Multiply both sides by 2 and rearranging further yields:

$$(\mathbf{x} - \boldsymbol{\mu}_2)^\top \mathbf{C}_2^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) - (\mathbf{x} - \boldsymbol{\mu}_1)^\top \mathbf{C}_1^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) + \ln \frac{|\mathbf{C}_2|}{|\mathbf{C}_1|} - 2 \ln \frac{\pi_2}{\pi_1} = 0.$$

Expanding the quadratic forms, we have:

$$\mathbf{x}^\top (\mathbf{C}_2^{-1} - \mathbf{C}_1^{-1}) \mathbf{x} - 2(\boldsymbol{\mu}_2^\top \mathbf{C}_2^{-1} - \boldsymbol{\mu}_1^\top \mathbf{C}_1^{-1}) \mathbf{x} + \boldsymbol{\mu}_2^\top \mathbf{C}_2^{-1} \boldsymbol{\mu}_2 - \boldsymbol{\mu}_1^\top \mathbf{C}_1^{-1} \boldsymbol{\mu}_1 + \ln \frac{|\mathbf{C}_2|}{|\mathbf{C}_1|} - 2 \ln \frac{\pi_2}{\pi_1} = 0.$$

- If $\mathbf{C}_1 \neq \mathbf{C}_2$, this is a **quadratic surface** in $\mathbb{R}^d$. - If $\mathbf{C}_1 = \mathbf{C}_2 = \mathbf{C}$, the quadratic term cancels and the boundary is linear:

$$(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \mathbf{C}^{-1} \mathbf{x} = \frac{1}{2} (\boldsymbol{\mu}_1^\top \mathbf{C}^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^\top \mathbf{C}^{-1} \boldsymbol{\mu}_2) + \ln \frac{\pi_1}{\pi_2}.$$

The decision boundary between two multivariate Gaussian classes depends on the means $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$, covariances $\mathbf{C}_1, \mathbf{C}_2$, and priors π_1, π_2 . It is in general, a quadratic boundary. In case, the covariances are equal, we obtain a linear hyperplane.

Now consider both classes

$$\text{Class 1: } \mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_1, \mathbf{C}_1), \quad P(\text{Class 1}) = \pi_1,$$

$$\text{Class 2: } \mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_2, \mathbf{C}_2), \quad P(\text{Class 2}) = \pi_2,$$

where the covariance matrices are diagonal:

$$\mathbf{C}_1 = \text{diag}(\sigma_{11}^2, \dots, \sigma_{1d}^2), \quad \mathbf{C}_2 = \text{diag}(\sigma_{21}^2, \dots, \sigma_{2d}^2).$$

The Bayes decision boundary satisfies

$$(\mathbf{x} - \boldsymbol{\mu}_2)^\top \mathbf{C}_2^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) - (\mathbf{x} - \boldsymbol{\mu}_1)^\top \mathbf{C}_1^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) + \ln \frac{|\mathbf{C}_2|}{|\mathbf{C}_1|} - 2 \ln \frac{\pi_2}{\pi_1} = 0.$$

Since $\mathbf{C}_1^{-1}$ and $\mathbf{C}_2^{-1}$ are diagonal, the quadratic term becomes

$$\mathbf{x}^\top (\mathbf{C}_2^{-1} - \mathbf{C}_1^{-1}) \mathbf{x} = \sum_{i=1}^d \left(\frac{1}{\sigma_{2i}^2} - \frac{1}{\sigma_{1i}^2} \right) x_i^2,$$

and the linear term is

$$-2(\boldsymbol{\mu}_2^\top \mathbf{C}_2^{-1} - \boldsymbol{\mu}_1^\top \mathbf{C}_1^{-1}) \mathbf{x} = -2 \sum_{i=1}^d \left(\frac{\mu_{2i}}{\sigma_{2i}^2} - \frac{\mu_{1i}}{\sigma_{1i}^2} \right) x_i.$$

The constant term is

$$\boldsymbol{\mu}_2^\top \mathbf{C}_2^{-1} \boldsymbol{\mu}_2 - \boldsymbol{\mu}_1^\top \mathbf{C}_1^{-1} \boldsymbol{\mu}_1 + \ln \frac{|\mathbf{C}_2|}{|\mathbf{C}_1|} - 2 \ln \frac{\pi_2}{\pi_1}.$$

Hence the decision boundary is axis-aligned:

$$\sum_{i=1}^d \left(\frac{1}{\sigma_{2i}^2} - \frac{1}{\sigma_{1i}^2} \right) x_i^2 - 2 \sum_{i=1}^d \left(\frac{\mu_{2i}}{\sigma_{2i}^2} - \frac{\mu_{1i}}{\sigma_{1i}^2} \right) x_i + \text{constant} = 0.$$

If $\mathbf{C}_1 = \mathbf{C}_2 = \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$, the quadratic term cancels and the boundary reduces to a linear hyperplane:

$$\sum_{i=1}^d \frac{\mu_{1i} - \mu_{2i}}{\sigma_i^2} x_i = \frac{1}{2} \sum_{i=1}^d \frac{\mu_{1i}^2 - \mu_{2i}^2}{\sigma_i^2} + \ln \frac{\pi_1}{\pi_2}.$$

If we have diagonal and unequal covariances, we get decision boundaries that are quadratic and axis-aligned quadratic boundary. If the covariance are equal, we get a linear hyperplane boundary.

2. Let $\mathbf{x} \in \mathbb{R}^d$ be a multivariate normal random vector. Derive the PDF of any k -element sub-vector of $\mathbf{x}$ with $k < d$.

Solution:

Let $\mathbf{X} \in \mathbb{R}^d$ be a random vector such that

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C}),$$

where $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\mathbf{C}$ is a $d \times d$ positive definite covariance matrix.

Partition $\mathbf{X}$, $\boldsymbol{\mu}$, and $\mathbf{C}$ as

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{pmatrix},$$

where $\mathbf{X}_1, \boldsymbol{\mu}_1 \in \mathbb{R}^k$ and $\mathbf{X}_2, \boldsymbol{\mu}_2 \in \mathbb{R}^{d-k}$.

The joint PDF of $\mathbf{X}$ is

$$f_{\mathbf{X}}(\mathbf{x}) = (2\pi)^{-d/2} |\mathbf{C}|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{C}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right).$$

Let $\mathbf{M} = \mathbf{C}^{-1}$ and partition it as

$$\mathbf{M} = \begin{pmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{pmatrix}, \quad \mathbf{M}_{21} = \mathbf{M}_{12}^\top.$$

Then

$$(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{M}(\mathbf{x} - \boldsymbol{\mu}) = (\mathbf{x}_1 - \boldsymbol{\mu}_1)^\top \mathbf{M}_{11}(\mathbf{x}_1 - \boldsymbol{\mu}_1) + 2(\mathbf{x}_1 - \boldsymbol{\mu}_1)^\top \mathbf{M}_{12}(\mathbf{x}_2 - \boldsymbol{\mu}_2) + (\mathbf{x}_2 - \boldsymbol{\mu}_2)^\top \mathbf{M}_{22}(\mathbf{x}_2 - \boldsymbol{\mu}_2).$$

We now marginalize over $\mathbf{x}_2$. The marginal density of $\mathbf{X}_1$ is

$$f_{\mathbf{X}_1}(\mathbf{x}_1) = \int_{\mathbb{R}^{d-k}} f_{\mathbf{X}} \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} d\mathbf{x}_2 = (2\pi)^{-d/2} |\mathbf{C}|^{-1/2} \int \exp\left(-\frac{1}{2}Q(\mathbf{x}_2)\right) d\mathbf{x}_2,$$

where

$$Q(\mathbf{x}_2) = (\mathbf{x}_2 - \boldsymbol{\mu}_2)^\top \mathbf{M}_{22}(\mathbf{x}_2 - \boldsymbol{\mu}_2) + 2(\mathbf{x}_1 - \boldsymbol{\mu}_1)^\top \mathbf{M}_{12}(\mathbf{x}_2 - \boldsymbol{\mu}_2) + (\mathbf{x}_1 - \boldsymbol{\mu}_1)^\top \mathbf{M}_{11}(\mathbf{x}_1 - \boldsymbol{\mu}_1).$$

We now complete the square in $\mathbf{x}_2$:

$$\begin{aligned} Q(\mathbf{x}_2) &= (\mathbf{x}_2 - \boldsymbol{\mu}_2 + \mathbf{M}_{22}^{-1} \mathbf{M}_{21}(\mathbf{x}_1 - \boldsymbol{\mu}_1))^\top \mathbf{M}_{22}(\mathbf{x}_2 - \boldsymbol{\mu}_2 + \mathbf{M}_{22}^{-1} \mathbf{M}_{21}(\mathbf{x}_1 - \boldsymbol{\mu}_1)) \\ &\quad + (\mathbf{x}_1 - \boldsymbol{\mu}_1)^\top (\mathbf{M}_{11} - \mathbf{M}_{12} \mathbf{M}_{22}^{-1} \mathbf{M}_{21})(\mathbf{x}_1 - \boldsymbol{\mu}_1). \end{aligned}$$

We now need to integrate out $\mathbf{x}_2$. The Gaussian integral

$$\int_{\mathbb{R}^{d-k}} \exp\left(-\frac{1}{2} \mathbf{u}^\top \mathbf{M}_{22} \mathbf{u}\right) d\mathbf{u} = (2\pi)^{(d-k)/2} |\mathbf{M}_{22}|^{-1/2}$$

yields

$$f_{\mathbf{X}_1}(\mathbf{x}_1) = (2\pi)^{-d/2} |\mathbf{C}|^{-1/2} (2\pi)^{(d-k)/2} |\mathbf{M}_{22}|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x}_1 - \boldsymbol{\mu}_1)^\top (\mathbf{M}_{11} - \mathbf{M}_{12} \mathbf{M}_{22}^{-1} \mathbf{M}_{21})(\mathbf{x}_1 - \boldsymbol{\mu}_1)\right).$$

We need to simplify using Schur complement identities. From matrix algebra, we have the following: [You can explain these steps to the students, but tell them that I am not expecting them to remember the Schur complement identities for the exam. In fact, there is a much simpler version of this entire derivation.](#)

$$\mathbf{M}_{11} - \mathbf{M}_{12} \mathbf{M}_{22}^{-1} \mathbf{M}_{21} = \mathbf{C}_{11}^{-1}, \quad |\mathbf{C}| |\mathbf{M}_{22}| = |\mathbf{C}_{11}|.$$

Substituting these into the expression for $f_{\mathbf{X}_1}(\mathbf{x}_1)$ gives

$$f_{\mathbf{X}_1}(\mathbf{x}_1) = (2\pi)^{-k/2} |\mathbf{C}_{11}|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x}_1 - \boldsymbol{\mu}_1)^\top \mathbf{C}_{11}^{-1}(\mathbf{x}_1 - \boldsymbol{\mu}_1)\right).$$

The marginal distribution of the k -dimensional subvector $\mathbf{X}_1$ is again multivariate normal:

$$\boxed{\mathbf{X}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \mathbf{C}_{11})}.$$

Shorter method that bypasses integrals: Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$ with $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\mathbf{C} \in \mathbb{R}^{d \times d}$ positive definite. By a matrix square root (for example the Cholesky factor) there exists a full-rank matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$ such that

$$\mathbf{C} = \mathbf{A}\mathbf{A}^\top.$$

Write $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ for a standard normal vector and represent

$$\mathbf{X} = \boldsymbol{\mu} + \mathbf{A}\mathbf{Z}.$$

Partition vectors and matrices conformably so that the first k components of $\mathbf{X}$ form the subvector of interest:

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{pmatrix},$$

where $\mathbf{X}_1, \boldsymbol{\mu}_1 \in \mathbb{R}^k$, $\mathbf{X}_2, \boldsymbol{\mu}_2 \in \mathbb{R}^{d-k}$, and $\mathbf{A}_1 \in \mathbb{R}^{k \times d}$ consists of the first k rows of $\mathbf{A}$.

Then the subvector is

$$\mathbf{X}_1 = \boldsymbol{\mu}_1 + \mathbf{A}_1\mathbf{Z}.$$

Because $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ and $\mathbf{A}_1$ is deterministic and linear, $\mathbf{X}_1$ is Gaussian (the linear combination of Gaussian random variables is Gaussian). This can be proved easily using moment generating functions.). $\mathbf{X}_1$ has the following mean and covariance

$$\mathbb{E}[\mathbf{X}_1] = \boldsymbol{\mu}_1, \quad \text{Cov}(\mathbf{X}_1) = \mathbf{A}_1 \text{Cov}(\mathbf{Z}) \mathbf{A}_1^\top = \mathbf{A}_1 \mathbf{A}_1^\top.$$

But by partitioning $\mathbf{C} = \mathbf{A}\mathbf{A}^\top$ we have

$$\mathbf{C} = \begin{pmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \end{pmatrix} \begin{pmatrix} \mathbf{A}_1^\top & \mathbf{A}_2^\top \end{pmatrix} = \begin{pmatrix} \mathbf{A}_1 \mathbf{A}_1^\top & \mathbf{A}_1 \mathbf{A}_2^\top \\ \mathbf{A}_2 \mathbf{A}_1^\top & \mathbf{A}_2 \mathbf{A}_2^\top \end{pmatrix}.$$

Hence the top-left block of $\mathbf{C}$ equals $\mathbf{A}_1 \mathbf{A}_1^\top$, i.e.

$$\text{Cov}(\mathbf{X}_1) = \mathbf{C}_{11}.$$

Therefore the marginal law of the k -dimensional subvector $\mathbf{X}_1$ is

$$\mathbf{X}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1, \mathbf{C}_{11}),$$

and its PDF is

$$f_{\mathbf{X}_1}(\mathbf{x}_1) = (2\pi)^{-k/2} |\mathbf{C}_{11}|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x}_1 - \boldsymbol{\mu}_1)^\top \mathbf{C}_{11}^{-1}(\mathbf{x}_1 - \boldsymbol{\mu}_1)\right).$$

3. Derive the moment generating function for a multivariate normal distribution.

Solution:

Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$, where $\boldsymbol{\mu} \in \mathbb{R}^d$ and $\mathbf{C} \in \mathbb{R}^{d \times d}$ is positive definite. In the derivation that follows, $\mathbf{t}$ is a parameter vector, replacing the scalar MGF parameter for scalar-valued random variables.

We will again represent $\mathbf{X}$ as a linear transformation of standard normal random variable. Let $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, and choose a matrix $\mathbf{A}$ such that $\mathbf{C} = \mathbf{A}\mathbf{A}^\top$ (for example, a Cholesky factor). Then we can write

$$\mathbf{X} = \boldsymbol{\mu} + \mathbf{A}\mathbf{Z}.$$

By definition of the MGF, we have:

$$M_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}[e^{\mathbf{t}^\top \mathbf{X}}] = \mathbb{E}[e^{\mathbf{t}^\top (\boldsymbol{\mu} + \mathbf{A}\mathbf{Z})}] = e^{\mathbf{t}^\top \boldsymbol{\mu}} \mathbb{E}[e^{(\mathbf{A}^\top \mathbf{t})^\top \mathbf{Z}}].$$

For a standard normal vector $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ (see below for the complete derivation for this case),

$$\mathbb{E}[e^{\mathbf{u}^\top \mathbf{Z}}] = \exp\left(\frac{1}{2} \mathbf{u}^\top \mathbf{u}\right), \quad \forall \mathbf{u} \in \mathbb{R}^d.$$

Take $\mathbf{u} = \mathbf{A}^\top \mathbf{t}$ to get

$$\mathbb{E}[e^{(\mathbf{A}^\top \mathbf{t})^\top \mathbf{Z}}] = \exp\left(\frac{1}{2} \mathbf{t}^\top \mathbf{A} \mathbf{A}^\top \mathbf{t}\right) = \exp\left(\frac{1}{2} \mathbf{t}^\top \mathbf{C} \mathbf{t}\right).$$

Hence the MGF of $\mathbf{X}$ is

$$\boxed{M_{\mathbf{X}}(\mathbf{t}) = \exp\left(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \mathbf{C} \mathbf{t}\right), \quad \mathbf{t} \in \mathbb{R}^d.}$$

Let $\mathbf{Z} \in \mathbb{R}^d$ be a standard multivariate normal vector:

$$\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d),$$

where $\mathbf{I}_d$ is the $d \times d$ identity matrix. The moment generating function (MGF) is defined as

$$M_{\mathbf{Z}}(\mathbf{t}) = \mathbb{E}[e^{\mathbf{t}^\top \mathbf{Z}}], \quad \mathbf{t} \in \mathbb{R}^d.$$

Since the components of $\mathbf{Z} = (Z_1, \dots, Z_d)^\top$ are independent standard normals:

$$\mathbb{E}[e^{\mathbf{t}^\top \mathbf{Z}}] = \mathbb{E}\left[e^{\sum_{i=1}^d t_i Z_i}\right] = \prod_{i=1}^d \mathbb{E}[e^{t_i Z_i}].$$

For a scalar $Z_i \sim \mathcal{N}(0, 1)$,

$$\mathbb{E}[e^{t_i Z_i}] = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{t_i z} e^{-z^2/2} dz.$$

Complete the square in the exponent:

$$t_i z - \frac{1}{2} z^2 = -\frac{1}{2} (z^2 - 2t_i z) = -\frac{1}{2} (z - t_i)^2 + \frac{1}{2} t_i^2.$$

Hence,

$$\mathbb{E}[e^{t_i Z_i}] = e^{t_i^2/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(z-t_i)^2/2} dz = e^{t_i^2/2}.$$

$$M_{\mathbf{Z}}(\mathbf{t}) = \prod_{i=1}^d e^{t_i^2/2} = e^{\frac{1}{2} \sum_{i=1}^d t_i^2} = e^{\frac{1}{2} \mathbf{t}^\top \mathbf{t}}.$$

The moment generating function of a d -dimensional standard normal vector is

$$\boxed{M_{\mathbf{Z}}(\mathbf{t}) = \exp\left(\frac{1}{2} \mathbf{t}^\top \mathbf{t}\right), \quad \mathbf{t} \in \mathbb{R}^d.}$$

4. Derive the differential entropy of a multivariate Gaussian random variable $\mathbf{X}$. This given as $H(\mathbf{X}) = E(-\log f_{\mathbf{X}}(\mathbf{x}))$

Solution: Let $\mathbf{X} \in \mathbb{R}^d$ be a multivariate Gaussian random vector:

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C}),$$

with mean $\boldsymbol{\mu} \in \mathbb{R}^d$ and positive definite covariance matrix $\mathbf{C} \in \mathbb{R}^{d \times d}$.

The differential entropy of a continuous random vector $\mathbf{X}$ with PDF $f_{\mathbf{X}}(\mathbf{x})$ is

$$h(\mathbf{X}) = - \int_{\mathbb{R}^d} f_{\mathbf{X}}(\mathbf{x}) \log f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x}.$$

The PDF of a multivariate Gaussian is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\mathbf{C}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right).$$

Taking the logarithm, we now get:

$$\log f_{\mathbf{X}}(\mathbf{x}) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\mathbf{C}| - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu}).$$

We now determine the entropy:

$$\begin{aligned} h(\mathbf{X}) &= -\mathbb{E}[\log f_{\mathbf{X}}(\mathbf{X})] \\ &= -\mathbb{E} \left[-\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\mathbf{C}| - \frac{1}{2} (\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \right] \\ &= \frac{d}{2} \log(2\pi) + \frac{1}{2} \log |\mathbf{C}| + \frac{1}{2} \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{X} - \boldsymbol{\mu})]. \end{aligned}$$

Using the trace trick, we get:

$$(\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{X} - \boldsymbol{\mu}) = \text{tr} \left(\mathbf{C}^{-1} (\mathbf{X} - \boldsymbol{\mu}) (\mathbf{X} - \boldsymbol{\mu})^\top \right).$$

We now take the expectation:

$$\mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{X} - \boldsymbol{\mu})] = \text{tr}(\mathbf{C}^{-1} \mathbf{C}) = d.$$

Thus, the differential entropy of a d -dimensional Gaussian vector is

$$h(\mathbf{X}) = \frac{d}{2} \log(2\pi) + \frac{1}{2} \log |\mathbf{C}| + \frac{d}{2} = \frac{1}{2} \log ((2\pi e)^d |\mathbf{C}|).$$

5. This question derives the Box-Muller method for generating samples from a normal distribution (scalar-valued case). We have seen a method in class which takes the inverse of the CDF. Unfortunately, the inverse CDF does not exist in closed-form for the normal distribution, and so another method was developed.

Solution: We want to generate samples from a standard normal distribution:

$$Z \sim \mathcal{N}(0, 1)$$

using uniform random variables $U_1, U_2 \sim \text{Uniform}(0, 1)$.

Step 1: Consider two independent standard normal variables $Z_0, Z_1 \sim \mathcal{N}(0, 1)$. Their joint pdf is

$$f_{Z_0, Z_1}(z_0, z_1) = \frac{1}{2\pi} \exp \left(-\frac{z_0^2 + z_1^2}{2} \right).$$

Step 2: We now convert to polar coordinates. Define

$$\begin{cases} Z_0 = R \cos \Theta \\ Z_1 = R \sin \Theta \end{cases}, \quad R \geq 0, \Theta \in [0, 2\pi)$$

The Jacobian of the transformation is $|J| = R$. Thus the joint pdf in polar coordinates becomes

$$f_{R, \Theta}(r, \theta) = f_{Z_0, Z_1}(r \cos \theta, r \sin \theta) \cdot R = \frac{1}{2\pi} r \exp \left(-\frac{r^2}{2} \right).$$

Step 3: We now factorize the joint PDF which we see is the product of a uniform PDF from 0 to 2π , and another PDF which is similar (but not identical) to an exponential distribution.

$$f_{R,\Theta}(r, \theta) = \underbrace{re^{-r^2/2}}_{f_R(r)} \cdot \underbrace{\frac{1}{2\pi}}_{f_\Theta(\theta)}$$

Hence:

$$\Theta \sim \text{Uniform}(0, 2\pi), \quad f_R(r) = re^{-r^2/2}, \quad r \geq 0.$$

One can easily verify that this density integrates to 1 (over 0 to ∞). It is also non-negative and is therefore a valid PDF.

Step 4: We now draw samples for the angle Θ and the radius R . It is easy for Θ :

$$\Theta = 2\pi U_2, \quad U_2 \sim \text{Uniform}(0, 1)$$

For R , we use the inverse CDF method, because an easy closed-form formula exists for $f_R(r)$:

$$F_R(r) = \int_0^r se^{-s^2/2} ds = 1 - e^{-r^2/2} = U_1 \quad \implies \quad R = \sqrt{-2 \ln U_1}.$$

Step 5: This is called the Box-Muller Transform. Finally, the standard normal samples are

$$\begin{cases} Z_0 = R \cos \Theta = \sqrt{-2 \ln U_1} \cos(2\pi U_2) \\ Z_1 = R \sin \Theta = \sqrt{-2 \ln U_1} \sin(2\pi U_2) \end{cases}.$$

It is interesting to note that there is no known closed-form method for generating one normal sample at random. But we have a method which generates two at random!

For an arbitrary mean and variance, this method is still applicable in the following way:

If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$X = \sigma Z + \mu.$$

This methods also extends to many other distributions:

- (a) Chi-squared: $\chi_k^2 = \sum_{i=1}^k Z_i^2$
- (b) Exponential: $X \sim \text{Exp}(\lambda)$ can be obtained via $X = \frac{1}{2}R^2$
- (c) Log-normal: $X = e^{\mu + \sigma Z}$